

7. The line l_1 has equation

$$\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-4}{3}$$

The line l_2 has equation

$$\mathbf{r} = \mathbf{i} + 3\mathbf{k} + t(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$$

where t is a scalar parameter.

(a) Show that l_1 and l_2 lie in the same plane.

(3)

(b) Write down a vector equation for the plane containing l_1 and l_2

(1)

(c) Find, to the nearest degree, the acute angle between l_1 and l_2

(3)

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4. The plane Π_1 has equation

$$\mathbf{r} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) + \mu(-\mathbf{i} + 2\mathbf{j} + \mathbf{k})$$

where λ and μ are scalar parameters.

- (a) Find a Cartesian equation for Π_1 (4)

The line l has equation

$$\frac{x-1}{5} = \frac{y-3}{-3} = \frac{z+2}{4}$$

- (b) Find the coordinates of the point of intersection of l with Π_1 (3)

The plane Π_2 has equation

$$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) = 5$$

- (c) Find, to the nearest degree, the acute angle between Π_1 and Π_2 (2)

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7. The line l_1 has equation

$$\mathbf{r} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} - 4\mathbf{k})$$

and the line l_2 has equation

$$\mathbf{r} = 5\mathbf{i} + p\mathbf{j} - 7\mathbf{k} + \mu(6\mathbf{i} + \mathbf{j} + 8\mathbf{k})$$

where λ and μ are scalar parameters and p is a constant.

The plane Π contains l_1 and l_2

- (a) Show that the vector $3\mathbf{i} - 10\mathbf{j} - \mathbf{k}$ is perpendicular to Π (2)
- (b) Hence determine a Cartesian equation of Π (2)
- (c) Hence determine the value of p (2)

Given that

- the lines l_1 and l_2 intersect at the point A
 - the point B has coordinates $(12, -11, 6)$
- (d) determine, to the nearest degree, the acute angle between AB and Π (4)



7. The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

where λ and μ are scalar parameters.

(a) Show that vector $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ is perpendicular to Π .

(2)

(b) Hence find a Cartesian equation of Π .

(2)

The line l has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 6 \\ -3 \end{pmatrix}$$

where t is a scalar parameter.

The point A lies on l .

Given that the shortest distance between A and Π is $2\sqrt{29}$

(c) determine the possible coordinates of A .

(4)

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8. Two birds are flying towards their nest, which is in a tree.

Relative to a fixed origin, the flight path of each bird is modelled by a straight line.

In the model, the equation for the flight path of the first bird is

$$\mathbf{r}_1 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ a \\ 0 \end{pmatrix}$$

and the equation for the flight path of the second bird is

$$\mathbf{r}_2 = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

where λ and μ are scalar parameters and a is a constant.

In the model, the angle between the birds' flight paths is 120°

- (a) Determine the value of a .

(4)

- (b) Verify that, according to the model, there is a common point on the flight paths of the two birds and find the coordinates of this common point.

(5)

The position of the nest is modelled as being at this common point.

The tree containing the nest is in a park.

The ground level of the park is modelled by the plane with equation

$$2x - 3y + z = 2$$

- (c) Hence determine the shortest distance from the nest to the ground level of the park.

(3)

- (d) By considering the model, comment on whether your answer to part (c) is reliable, giving a reason for your answer.

(1)



5. The line l_1 has equation $\frac{x+5}{1} = \frac{y+4}{-3} = \frac{z-3}{5}$

The plane Π_1 has equation $2x + 3y - 2z = 6$

(a) Find the point of intersection of l_1 and Π_1 (2)

The line l_2 is the reflection of the line l_1 in the plane Π_1

(b) Show that a vector equation for the line l_2 is

$$\mathbf{r} = \begin{pmatrix} -7 \\ 2 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 6 \\ 2 \end{pmatrix}$$

where μ is a scalar parameter. (5)

The plane Π_2 contains the line l_1 and the line l_2

(c) Determine a vector equation for the line of intersection of Π_1 and Π_2 (2)

The plane Π_3 has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ a \end{pmatrix} = b$ where a and b are constants.

Given that the planes Π_1 , Π_2 and Π_3 form a sheaf,

(d) determine the value of a and the value of b . (3)

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